

GIWON SONG

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EDUCATION

Seoul National University

Bachelor of Science in Electronic and Computer Engineering — 2029 (expected)

- Cumulative GPA: 4.04 / 4.3
- Academic Excellence Scholarship (Merit-based) : 2023 – 2024

Korea Science Academy of KAIST (KSA)

High School Diploma — 2023

RESEARCH EXPERIENCE

Quantum Information and Quantum Computer Lab (QuIQCL), Seoul National University

Undergraduate Research Intern — July 2024 – Apr 2025

- Conducted research on the theoretical limits of quantum computing and algorithmic efficiency.

Undergraduate Research Program (URP) Participant — July 2023 – Aug 2023

- Co-developed IQUIP (Ion trap Quantum computing User Interface Project). Project Repository : [\[GitHub\]](#)

Korea Science Academy of KAIST (KSA)

Student Researcher — 2021

- Researched advanced implementations of Neural Radiance Fields (NeRF) for multiple objects.

PUBLICATIONS

Manuscript in Preparation

- Hwang, Y.^{*}, **Song, G.^{*}**, Park, K., Kwon, H.[†], Kim, T.[†] “Efficient expectation value estimation for quantum circuits via extended stabilizer frameworks and adaptive variance estimation” (Manuscript in preparation).

**These authors contributed equally to this work. [†]Corresponding authors.*

Publications

- Park, K., Choi, K., Kim, M., **Song, G.**, Kim, T.[†] “Quantum Linear Multistep Method for Using a Quantum Oracle with Differential Equations.” ArXiv preprint, 2025. ArXiv : [\[ArXiv\]](#)

**These authors contributed equally to this work. [†]Corresponding author.*

AWARDS & HONORS

- **Special Prize (IBM Quantum)** | Korea Q Hackathon | 2024
- **Advanced Badge (Completed all exercises)** | IBM Quantum Spring Challenge | 2024
- **Director's Award, Quantum Information Research Support Center** | Korea Q Hackathon | 2023
- **Silver Prize** | Korea Olympiad in Informatics (KOI) | 2016

EXTRACURRICULAR ACTIVITIES

Republic of Korea Army (ROKA)

Cryptanalyst — May 2025 – Nov 2026

- Decrypted complex ciphers by training a large language model (LLM) targeted to the specific properties of the plaintext.

SKILLS & LANGUAGES

Technical skills : C++, Python, Qiskit, LaTeX, Git

Languages : Korean (native), English (professional working proficiency), Japanese (professional working proficiency)

- iBT TOEFL: 5.5/6 (110/120)
- JLPT: N1(Results Expected August 2026), N2 (165/180)